1 I

1.1 (1)

From the continuity of the wave function at $x = 0$,

$$\boxed{1 + r = t}.$$

1.2 (2)

Integrating the Schrödinger equation across the delta potential gives

$$\phi_2'(0) - \phi_1'(0) = -\frac{2mV_0}{\hbar^2}\phi(0).$$

Substituting the wave functions, we obtain

$$\boxed{t + r = \frac{2i}{\omega}t + 1}.$$

1.3 (3)

Combining the results of (1) and (2),

$$r = \frac{i}{\omega}t, \qquad t = \frac{1}{1 - i/\omega}.$$

Thus, the reflection and transmission probabilities are

$$\boxed{|r|^2 = \frac{1}{1 + \omega^2}},$$

and

$$\boxed{|t|^2 = \frac{\omega^2}{1 + \omega^2}}.$$

1.4 (4)

For a fixed V_0, since $\omega \propto \sqrt{E}$, we obtain

$$\begin{aligned} E \to 0: &\quad |r|^2 \to 1, \qquad |t|^2 \to 0, \\ E \to \infty: &\quad |r|^2 \to 0, \qquad |t|^2 \to 1. \end{aligned}$$

When $V_0 \to -V_0$, the sign of ω is reversed.

However, since both probabilities depend only on ω^2, the reflection and transmission probabilities remain unchanged for any fixed positive energy.

2 II

2.1 (5)

Since the delta potential is symmetric about $x = 0$, the reflection and transmission amplitudes are the same for incidence from either side.

By the superposition principle, the reflected and transmitted waves from the two incident waves can be added.

Thus,

$$\boxed{\begin{aligned} B_1 &= rA_1 + tD_1, \\ C_1 &= tA_1 + rD_1. \end{aligned}}$$

2.2 (6)

Rearranging the equations in (5),

$$\begin{aligned} C_1 &= \left(t - \frac{r^2}{t}\right) A_1 + \frac{r}{t} B_1, \\ D_1 &= -\frac{r}{t} A_1 + \frac{1}{t} B_1. \end{aligned}$$

Therefore,

$$\begin{pmatrix} C_1 \\ D_1 \end{pmatrix} = Z \begin{pmatrix} A_1 \\ B_1 \end{pmatrix},$$

where

$$\boxed{Z = \begin{pmatrix} t - \dfrac{r^2}{t} & \dfrac{r}{t} \\ -\dfrac{r}{t} & \dfrac{1}{t} \end{pmatrix}}.$$

2.3 (7)

Comparing the two expressions for $\psi_2(x)$, we obtain

$$\begin{pmatrix} A_2 \\ B_2 \end{pmatrix} = Y \begin{pmatrix} C_1 \\ D_1 \end{pmatrix},$$

where

$$\boxed{Y = \begin{pmatrix} e^{iqa} & 0 \\ 0 & e^{-iqa} \end{pmatrix}}.$$

2.4 (8)

Applying the same transfer matrix to the second delta potential,

$$\begin{pmatrix} C_2 \\ D_2 \end{pmatrix} = Z \begin{pmatrix} A_2 \\ B_2 \end{pmatrix}.$$

Combining this with the results of (6) and (7), we obtain

$$\boxed{X = ZYZ}.$$

2.5 (9)

When $a = 2n\pi/q$, we have

$$Y = \mathbb{I}_2,$$

and thus

$$X = Z^2.$$

From the results of Part I,

$$\frac{r}{t} = \frac{i}{\omega}, \qquad \frac{1}{t} = 1 - \frac{i}{\omega}.$$

Therefore,

$$Z = \begin{pmatrix} 1 + \dfrac{i}{\omega} & \dfrac{i}{\omega} \\ -\dfrac{i}{\omega} & 1 - \dfrac{i}{\omega} \end{pmatrix}.$$

Squaring the matrix, we obtain

$$Z^2 = \begin{pmatrix} 1 + \dfrac{2i}{\omega} & \dfrac{2i}{\omega} \\ -\dfrac{2i}{\omega} & 1 - \dfrac{2i}{\omega} \end{pmatrix}.$$

Using the boundary conditions $A_1 = 1$, $B_1 = r_2$, $C_2 = t_2$, and $D_2 = 0$,

$$\begin{pmatrix} t_2 \\ 0 \end{pmatrix} = Z^2 \begin{pmatrix} 1 \\ r_2 \end{pmatrix}.$$

Solving for r_2 and t_2,

$$r_2 = \frac{2i}{\omega - 2i}, \qquad t_2 = \frac{\omega}{\omega - 2i}.$$

Thus,

$$\boxed{|r_2|^2 = \frac{4}{\omega^2 + 4}},$$

and

$$\boxed{|t_2|^2 = \frac{\omega^2}{\omega^2 + 4}}.$$

Compared with the results of (3), these expressions are equivalent to those for a single delta potential with twice the original strength.

Thus, under the given condition, the two delta potentials effectively act as a single stronger potential in terms of reflection and transmission probabilities.